

Continuity and the Intermediate Value Theorem

MATH 1334

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Section 2.4: continuity

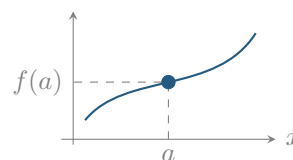
Definition. A function f is *continuous at a* if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Check three things:

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a} f(x)$ exists as a finite number.
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

Graphically: the graph has no break at a .



Algebraically: the plug-in rule works at a .



Example 1. Is f continuous at $a = -\frac{1}{2}$?

$$f(x) = \begin{cases} \frac{6x^2 - x - 2}{2x + 1}, & x \neq -\frac{1}{2}, \\ -3, & x = -\frac{1}{2}. \end{cases}$$

Check the value and the limit. We have $f(-\frac{1}{2}) = -3$. For $x \neq -\frac{1}{2}$,

$$\frac{6x^2 - x - 2}{2x + 1} = \frac{(2x + 1)(3x - 2)}{2x + 1} = 3x - 2.$$

Hence

$$\lim_{x \rightarrow -1/2} f(x) = 3\left(-\frac{1}{2}\right) - 2 = -\frac{7}{2} \neq -3.$$

Therefore f is not continuous at $x = -\frac{1}{2}$.

Corollary. *Polynomials and rational functions are continuous on their domains.*

Proof. Immediate from the plug-in rule. □

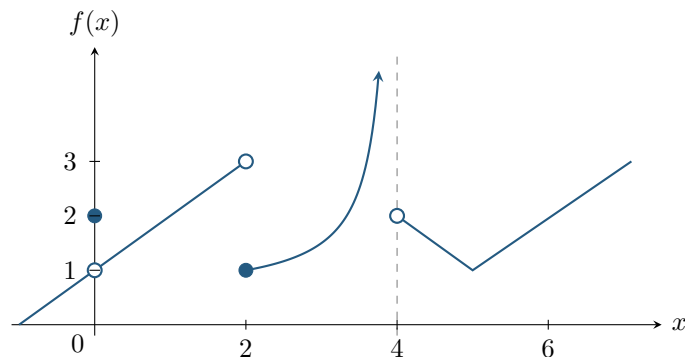
Types of discontinuities

Definition. A *removable discontinuity* at a occurs when the finite limit $\lim_{x \rightarrow a} f(x)$ exists, but $f(a)$ is undefined or differs from that limit.

Definition. A *jump discontinuity* at a occurs when the two one-sided limits are finite and unequal.

Definition. An *infinite discontinuity* at a occurs when at least one one-sided limit is $+\infty$ or $-\infty$.

Example 2. Classify the discontinuities in the graph.



x	Type
0	Removable
2	Jump
4	Infinite

One-sided continuity

Definition. A function is *right-continuous* at a if $\lim_{x \rightarrow a^+} f(x) = f(a)$, and *left-continuous* at a if $\lim_{x \rightarrow a^-} f(x) = f(a)$.

A function is continuous on an interval when it is continuous at every interior point (a point that is not an endpoint), with right-continuity at an included left endpoint and left-continuity at an included right endpoint.

Example 3. The function $f(x) = \sqrt{x}$ is continuous on $[0, \infty)$.

For $a > 0$, the root law gives $\lim_{x \rightarrow a} \sqrt{x} = \sqrt{a}$. At the endpoint,

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0 = f(0),$$

so f is right-continuous at 0.

Problems: choosing constants for continuity

1. Find every $k \leq 5$ such that H is continuous on $(-\infty, 5]$:

$$H(x) = \begin{cases} 3x + 1, & x < k, \\ x - 3, & k \leq x \leq 5. \end{cases}$$

Match the pieces. Each formula is continuous on its interval. At the joining point,

$$\lim_{x \rightarrow k^-} H(x) = 3k + 1, \quad H(k) = k - 3.$$

For $k < 5$, the right-hand limit is also $k - 3$. Thus continuity requires

$$3k + 1 = k - 3,$$

$$2k = -4,$$

$$k = -2.$$

The endpoint case $k = 5$ would require the same equality for left-continuity, and does not satisfy it. With $k = -2$, the pieces match and H is also left-continuous at 5. Therefore the answer is $k = -2$.

2. Find all m and b that make P continuous everywhere:

$$P(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x < 2, \\ b, & x = 2, \\ mx - 1, & x > 2. \end{cases}$$

Match both limits and the value. The only joining point is $x = 2$.

$$\lim_{x \rightarrow 2^-} P(x) = \lim_{x \rightarrow 2^-} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2^-} (x + 2) = 4,$$

$$\lim_{x \rightarrow 2^+} P(x) = 2m - 1,$$

$$P(2) = b.$$

Continuity requires all three quantities to agree:

$$4 = 2m - 1 = b.$$

Consequently,

$$b = 4, \quad m = \frac{5}{2}.$$

With these choices, P is continuous at 2 and on each side of 2.

Operations that preserve continuity

Proposition. Suppose f and g are continuous on their domains. Then the following functions are continuous wherever they are defined:

$$f + g, \quad f - g, \quad cf, \quad fg, \quad \frac{f}{g}, \quad f^n, \quad \sqrt[n]{f},$$

where c is a constant and n is a positive integer. For a quotient, require $g(x) \neq 0$; for an even root, require $f(x) \geq 0$.

Proof. For the sum, at a point a in the common domain,

$$\begin{aligned} \lim_{x \rightarrow a} (f(x) + g(x)) &= \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) \\ &= f(a) + g(a) = (f + g)(a). \end{aligned}$$

The other statements follow from the corresponding limit laws, with limits taken within the domain. $\square$

Continuity and composition

Proposition. Suppose $\lim_{x \rightarrow a} g(x) = L$ and F is continuous at L . If $F(g(x))$ is defined for the nearby inputs under consideration, then

$$\lim_{x \rightarrow a} F(g(x)) = F\left(\lim_{x \rightarrow a} g(x)\right) = F(L).$$

In particular, if g is continuous at a and F is continuous at $g(a)$, then $F \circ g$ is continuous at a .

Example 4. Let $\lim_{x \rightarrow 0} g(x) = -1$ and $F(x) = x^2 - 2x + 1$.

Composition. Since F is a polynomial, it is continuous at -1 . Therefore

$$\begin{aligned} \lim_{x \rightarrow 0} (F \circ g)(x) &= \lim_{x \rightarrow 0} F(g(x)) \\ &= F\left(\lim_{x \rightarrow 0} g(x)\right) \\ &= F(-1) = 1 + 2 + 1 = 4. \end{aligned}$$

Continuity of the trigonometric functions

Theorem. *The functions $\sin x$, $\cos x$, and $\tan x$ are continuous on their domains.*

Lemma. *For angles measured in radians,*

$$\lim_{h \rightarrow 0} \sin h = 0, \quad \lim_{h \rightarrow 0} \cos h = 1.$$

Proof of the lemma. Using the limit $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$ established previously,

$$\lim_{h \rightarrow 0} \sin h = \lim_{h \rightarrow 0} \left(h \frac{\sin h}{h} \right) = 0 \cdot 1 = 0.$$

Near 0, cosine is positive. The identity $\cos^2 h = 1 - \sin^2 h$ and the root law give

$$\lim_{h \rightarrow 0} \cos h = \lim_{h \rightarrow 0} \sqrt{1 - \sin^2 h} = 1. \quad \square$$

Proof of the theorem.

(i) Sine. Fix $a \in \mathbb{R}$. Write $h = x - a$, so $h \rightarrow 0$ as $x \rightarrow a$. By the angle-addition identity,

$$\begin{aligned} \lim_{x \rightarrow a} \sin x &= \lim_{x \rightarrow a} \sin((x - a) + a) \\ &= \lim_{x \rightarrow a} (\sin(x - a) \cos a + \sin a \cos(x - a)) \\ &= \cos a \lim_{h \rightarrow 0} \sin h + \sin a \lim_{h \rightarrow 0} \cos h \\ &= \cos a \cdot 0 + \sin a \cdot 1 = \sin a. \end{aligned}$$

Thus sine is continuous at every real number a .

(ii) Cosine. Since $\cos x = \sin(x + \pi/2)$, sine continuity and composition give

$$\lim_{x \rightarrow a} \cos x = \lim_{x \rightarrow a} \sin(x + \pi/2) = \sin(a + \pi/2) = \cos a.$$

Thus cosine is continuous at every real number a .

(iii) Tangent. If a is in the domain of tangent, then $\cos a \neq 0$. By the quotient law,

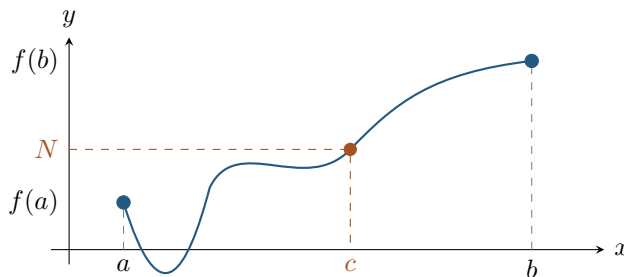
$$\lim_{x \rightarrow a} \tan x = \lim_{x \rightarrow a} \frac{\sin x}{\cos x} = \frac{\sin a}{\cos a} = \tan a.$$

Thus tangent is continuous wherever it is defined. $\square$

The Intermediate Value Theorem

Theorem (Intermediate Value Theorem). Suppose f is continuous on $[a, b]$. If N lies between $f(a)$ and $f(b)$, then there is a $c \in [a, b]$ such that

$$f(c) = N.$$



Idea: Continuous functions achieve their intermediate values.

Example 5. Does the equation $4x^3 - 6x^2 + 3x - 2 = 0$ have a solution?

Intermediate Value Theorem. Let $f(x) = 4x^3 - 6x^2 + 3x - 2$. Since f is a polynomial, it is continuous on $[0, 2]$. Also,

$$f(0) = -2, \quad f(2) = 32 - 24 + 6 - 2 = 12.$$

Because $-2 < 0 < 12$, the IVT guarantees a number $c \in (0, 2)$ with $f(c) = 0$. Thus the equation has a solution.

Example 6. You travel from home to campus from 7–8 a.m., then return home along the same path from 7–8 p.m. Must there be a clock time when you are at the same place during both trips?

Intermediate Value Theorem. Yes. Imagine the two trips happening simultaneously: one traveler starts at home and the other at campus. Their positions along the path vary continuously, and their order reverses by the end of the hour. The difference of their positions therefore changes sign. By the IVT, that difference must be zero sometime: they meet. This gives the same place at the same number of minutes after 7 on both trips.